

Quantum Harmonic Oscillator

The time-independent Schrödinger equation is

$$\hat{H}(\sigma) = E\sigma$$

here $\hat{H}(\sigma) = \underbrace{\frac{1}{2} m \omega^2 x^2}_{\text{"spring"}} \sigma - \frac{1}{2} \frac{\hbar^2}{m} \sigma'' = E\sigma$

$$\Rightarrow \sigma'' = \left(\frac{E - \frac{1}{2} m \omega^2 x^2}{-\frac{1}{2} \frac{\hbar^2}{m}} \right) \sigma = \frac{-2mE + m^2 \omega^2 x^2}{\hbar^2} \sigma$$

To simplify notation, let $r = \sqrt{\frac{\hbar}{m\omega}}$ so $r^2 = \frac{\hbar}{m\omega}$

and let $u = \frac{x}{r} \Rightarrow \frac{du}{dx} = \frac{1}{r} \Rightarrow \frac{d\sigma}{dx} = \frac{1}{r} \frac{d\sigma}{du} = \frac{d\sigma}{du} \frac{du}{dx}$

Thus $\sigma'' = \frac{1}{r^2} \frac{d^2\sigma}{du^2} = \left(\frac{x^2}{r^4} - \frac{2mE}{\hbar} \cdot \frac{1}{\hbar} \right) \sigma$

$$\Rightarrow \frac{d^2\sigma}{du^2} = \left(\frac{x^2}{r^2} - \frac{2mE}{\hbar} r^2 \frac{1}{\hbar} \right) \sigma = (u^2 - \frac{2E}{\omega\hbar}) \sigma$$

set $\epsilon = \frac{2E}{\omega\hbar} \Rightarrow \frac{d^2\sigma}{du^2} = (u^2 - \epsilon) \sigma$

We can stop here and note for $u \gg \epsilon$ $\sigma'' \sim u^2 \sigma$

$$\Rightarrow \sigma \approx c_1 e^{f(u)} + c_2 e^{-f(u)} \quad f \sim u^2$$

Instead we use this idea $\uparrow$ to get more specific information

Set $\sigma(u) = s(u) e^{-u^2/2}$ (normalization)

$$\Rightarrow \frac{d^2\sigma}{du^2} = \frac{d}{du} \left(\frac{ds}{du} e^{-u^2/2} - u s(u) e^{-u^2/2} \right)$$

$$= \frac{d^2s}{du^2} e^{-u^2/2} - u \frac{ds}{du} e^{-u^2/2} - \left[s(u) e^{-u^2/2} + \right.$$

$$\left. (u \frac{ds}{du} e^{-u^2/2} - u^2 s(u) e^{-u^2/2}) \right] = (u^2 - \epsilon) \sigma$$
$$= (u^2 - \epsilon) s e^{-u^2/2}$$

$$\Rightarrow e^{-u^2/2} \left[\frac{d^2 s}{du^2} - 2u \frac{ds}{du} + (\epsilon - 1)s \right] = 0$$

$$\Rightarrow s'' - 2us' + \lambda s = 0, \quad \lambda = \epsilon - 1$$

this is called Hermite's Equation, λ corresponds to the energy states of the oscillator.

Ex: As we will see, λ must be even, so suppose

$$\lambda = 4, \text{ then } s'' - 2us' + 4s = 0$$

We can see that $s(u) = 4u^2 - 2$ solves the equation.

$$\begin{aligned} s'' = 8, s' = 8u &\Rightarrow 8 - 2u(8u) + 4(4u^2 - 2) \\ &= 8 - 16u^2 + 16u^2 - 8 = 0 \quad \checkmark \end{aligned}$$

By reduction of order with $y_1 = 4u^2 - 2$:

$$y_2 = y_1 \int \frac{e^{-\int p(u) du}}{y_1^2} du = (4u^2 - 2) \int \frac{e^{-\int -2u du}}{(4u^2 - 2)^2} du$$

$$\Rightarrow y_2 = (4u^2 - 2) \int (4u^2 - 2)^{-2} e^{u^2} du$$

$\uparrow$
 undoes normalization.

$$\text{let } S = \sum_{n=0}^{\infty} a_n u^n, \quad S' = \sum_{n=1}^{\infty} n a_n u^{n-1}, \quad S'' = \sum_{n=2}^{\infty} n(n-1) a_n u^{n-2}$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1) a_n u^{n-2} - 2u \sum_{n=1}^{\infty} n a_n u^{n-1} + \lambda \sum_{n=0}^{\infty} a_n u^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} u^n + \sum_{n=1}^{\infty} -2n a_n u^n + \sum_{n=0}^{\infty} \lambda a_n u^n = 0$$

$$\Rightarrow (2a_2 + \lambda a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1) a_{n+2} - (2n - \lambda) a_n] u^n = 0$$

$$\Rightarrow \begin{cases} 2a_2 + \lambda a_0 = 0 \\ (n+2)(n+1) a_{n+2} - (2n - \lambda) a_n = 0 \Rightarrow a_{n+2} = \frac{2n - \lambda}{(n+2)(n+1)} a_n \end{cases}$$

Note that we expect for $n \gg 0$ $a_{n+2} \approx \frac{2n}{n^2} a_n \approx \frac{a_n}{n}$

$$\rightarrow a_i \approx \frac{C}{(\frac{n}{2})!} \text{ or } S = C e^{u^2} \text{ in } u \text{ (see Ex)}$$

which undoes normalization. So we must impose that the series is finite. This means for large

$$n \text{ we must have } 2n - \lambda = 0 \Rightarrow \lambda = 2n$$

$$\Rightarrow E - 1 = 2n \Rightarrow \frac{2E}{\hbar \omega} - 1 = 2n$$

$$\Rightarrow E = \frac{(2n+1)\hbar\omega}{2} = \hbar(n + \frac{1}{2})\omega$$

so $E_n = \hbar(n + \frac{1}{2})\omega$ guarantees a finite series.

These are the energy states.

If $n=0$: $\lambda=0$ or $\varepsilon=1$, so $a_2 = -\frac{\lambda}{2}a_0 = 0 \Rightarrow a_{\text{even}} = 0$
 and imposing odds zero, i.e. $a_1 = a_3 = a_5 = \dots = 0$ we get
 $s(u) = a_0 \Rightarrow \sigma = S e^{-u^2/2} = a_0 e^{-u^2/2}$

If $n=1$: $\lambda=2$, $\varepsilon=3 \Rightarrow a_3 = a_{1+2} = \frac{2-\lambda}{6} a_1 = 0$

and if $a_0 = a_2 = \dots = 0$ then

$$s(u) = a_1 u \Rightarrow \sigma = a_1 u e^{-u^2/2}$$

If $n=2$: $\lambda=4$, $\varepsilon=5 \Rightarrow a_4 = 0$, $a_2 = -\frac{\lambda}{2}a_0 = -2a_0$

and if $a_1 = a_3 = \dots = 0$ then

$$s(u) = a_0 + a_2 u^2 = a_0 - 2a_0 u^2 = (1-2u^2)a_0$$

if $a_0 = -2 \Rightarrow s(u) = 4u^2 - 2$ (example)

$$\Rightarrow \sigma = (1-2u^2)a_0 e^{-u^2/2}$$

If $n=3$: $\lambda=6$, $\varepsilon=7 \Rightarrow a_5 = \frac{2(3)-6}{20} a_3 = 0$

if $a_0 = a_2 = \dots = 0$ then

$$a_3 = \frac{2(1)-6}{6} a_1 = -\frac{4}{6} a_1 = -\frac{2}{3} a_1$$

so $s(u) = a_1 u + a_3 u^3 = a_1 u - \frac{2}{3} a_1 u^3 = (u - \frac{2}{3} u^3) a_1$

if $a_1 = -12 \Rightarrow s(u) = 8u^3 - 12u$, $\sigma(u) = (u - \frac{2}{3} u^3) a_1 e^{-u^2/2}$

These polynomials

n	$S(u)$	H_n	constant
0	$1 \cdot a_0$	1	$a_0 = 1$
1	$u \cdot a_1$	$2u$	$a_1 = 2$
2	$(1 - 2u^2) a_0$	$4u^2 - 2$	$a_0 = -2$
3	$(u - \frac{2}{3}u^3) a_1$	$8u^3 - 12u$	$a_1 = -12$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

we constant multiples of the Hermite polynomials
(hence the name of the equation).

The multiple they are off by is called the normalization factor. H_n is defined so the leading coefficient is a power of 2. So $H_n = 2^n u^n + \dots$

For each n our solution becomes $\sigma_n(u) = S_n(u) e^{-u^2/2}$
 $\Rightarrow \sigma_n(u) = \frac{C}{N(n)} H_n(u) e^{-u^2/2} = \frac{C}{N(n)} H_n\left(\frac{x}{r}\right) e^{-\frac{x^2}{2r^2}} = \sigma_n(x)$

we had $r = \sqrt{\frac{\hbar}{m\omega}}$ and so $\sigma_n(x) = \frac{C}{N(n)} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega x^2}{2\hbar}}$

It turns out $\frac{C}{N(n)} = \left(\frac{1}{2^n n! \sqrt{\pi \hbar}}\right)^{1/2} = \sqrt{\frac{1}{2^n n!}} \left(\frac{m\omega}{\pi \hbar}\right)^{1/4}$

and so the general energy states are

$$\sigma_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar}\right)^{1/4} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega x^2}{2\hbar}}, n=0,1,2,\dots$$

Laplacian $\Delta f = \nabla(\nabla f)$ (divergence of gradient) (Heat equation $f(x, y, z, t)$)
 $= f_{xx} + f_{yy} + f_{zz}$ $\Delta f = f_t$

Consider PDE $\Delta f = 0$ (Laplace's equation)

If we convert $x, y, z \rightarrow$ spherical (ρ, θ, ϕ) we obtain

$$\Delta f = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{\rho^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} = 0$$

If we assume $f = R(\rho)T(\theta)P(\phi)$ then we get

$$\Delta f = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial R}{\partial \rho} T P \right) + \frac{1}{\rho^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} R P \right) + \frac{1}{\rho^2 \sin^2 \theta} \frac{\partial^2 P}{\partial \phi^2} R T = 0$$

$$\Rightarrow \frac{1}{R} \frac{d}{d\rho} \left(\rho^2 \frac{dR}{d\rho} \right) + \frac{1}{T \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dT}{d\theta} \right) + \frac{1}{P \sin^2 \theta} \frac{d^2 P}{d\phi^2} = 0$$

Next assume the ϕ angle has no change in Δf i.e. $\frac{dP}{d\phi} = 0$

$$\Rightarrow \frac{1}{R} \frac{d}{d\rho} \left(\rho^2 \frac{dR}{d\rho} \right) + \frac{1}{T \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dT}{d\theta} \right) = 0$$

This process is called separation of variables. Now, we have an equation with two functions of separate variables, and so they must be constant to add to 0, i.e.

$$\begin{cases} \frac{1}{R} \frac{d}{d\rho} \left(\rho^2 \frac{dR}{d\rho} \right) = \lambda & \leftarrow \text{constant} \\ \frac{1}{T \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dT}{d\theta} \right) = -\lambda & \leftarrow \end{cases}$$

Consider θ equation (imagine distance ρ is fixed)

$$\frac{1}{T \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dT}{d\theta} \right) = -\lambda$$

$$\text{let } x = +\cos \theta \Rightarrow \frac{d}{d\theta} = \frac{dx}{d\theta} \frac{d}{dx} = -\sin \theta \frac{d}{dx}$$

$$\Rightarrow \frac{1}{\sin \theta} \frac{d}{d\theta} = -\frac{d}{dx} \quad \text{and} \quad \sin^2 \theta = \left(\sqrt{1 - \cos^2 \theta} \right)^2 = \left(\sqrt{1 - x^2} \right)^2 = 1 - x^2$$

$$\Rightarrow \frac{1}{T \sin \theta} \frac{d}{d\theta} \left(\sin \theta \cdot \frac{d}{d\theta} (T) \right) = -\lambda$$

$$\Rightarrow \frac{1}{\sin \theta} \cdot \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} (T) \right) = -\lambda T$$

$$\Rightarrow -\frac{d}{dx} \left(-\sin^2 \theta \frac{dT}{dx} \right) = -\lambda T$$

$$\Rightarrow \frac{d}{dx} \left((1-x^2) \frac{dT}{dx} \right) = -\lambda T$$

$$\Rightarrow (1-x^2) \frac{d^2 T}{dx^2} - 2x \frac{dT}{dx} + \lambda T = 0$$

switch $T \rightarrow y$ and let $\lambda = \alpha(\alpha+1)$ (Rodriguez's formula) ^{real}
(Legendre's Equation)

$$\Rightarrow (1-x^2) y'' - 2x y' + \alpha(\alpha+1) y = 0$$

Suppose $y = \sum_{n=0}^{\infty} a_n x^n$, then $(1-x^2)y'' - 2xy' + \alpha(\alpha+1)y = 0$

$$\Rightarrow (1-x^2) \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 2x \sum_{n=1}^{\infty} n a_n x^{n-1} + \alpha(\alpha+1) \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n x^n - \sum_{n=1}^{\infty} 2n a_n x^n + \sum_{n=0}^{\infty} \alpha(\alpha+1)a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n - \sum_{n=2}^{\infty} n(n-1)a_n x^n - \sum_{n=1}^{\infty} 2n a_n x^n + \sum_{n=0}^{\infty} \alpha(\alpha+1)a_n x^n = 0$$

$$\Rightarrow (2a_2 + \alpha(\alpha+1)a_0) + (6a_3 - 2a_1 + \alpha(\alpha+1)a_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)a_{n+2} - n(n-1)a_n - 2na_n + \alpha(\alpha+1)a_n] x^n = 0$$

$$\Rightarrow \begin{cases} 2a_2 + \alpha(\alpha+1)a_0 = 0 & (n=0) \\ 6a_3 - 2a_1 + \alpha(\alpha+1)a_1 = 0 & (n=1) \end{cases} \quad \boxed{a_{n+2} = - \frac{(\alpha-n)(\alpha+n+1)}{(n+1)(n+2)} a_n} \quad n \geq 2$$

$$(n+2)(n+1)a_{n+2} + [\alpha(\alpha+1) - n(n-1) - 2n]a_n = 0, \quad n \geq 2$$

evens $\alpha^2 + \alpha - n^2 - n = (\alpha-n)(\alpha+n) + (\alpha-n)$ odds
 $= (\alpha-n)(\alpha+n+1)$

$$n=0: \quad a_2 = - \frac{\alpha(\alpha+1)}{2} a_0$$

$$n=1: \quad a_3 = \frac{2-\alpha(\alpha+1)}{6} a_1 = \frac{2-\alpha^2-\alpha}{2 \cdot 3} a_1$$

$$\Rightarrow a_3 = - \frac{\alpha^2 + \alpha - 2}{2 \cdot 3} a_1 = - \frac{(\alpha+1)(\alpha-2)}{2 \cdot 3} a_1$$

$$n=2: \quad a_4 = - \frac{(\alpha-2)(\alpha+3)}{3 \cdot 4} a_2$$

$$\Rightarrow a_4 = (-1)^2 \frac{\alpha(\alpha-2)(\alpha+1)(\alpha+3)}{4!} a_0$$

$$n=3: \quad a_5 = - \frac{(\alpha-3)(\alpha+4)}{4 \cdot 5} a_3$$

$$\Rightarrow a_5 = (-1)^2 \frac{(\alpha-1)(\alpha-3)(\alpha+2)(\alpha+4)}{5!} a_1$$

evens: $a_{2n} = (-1)^n \frac{\alpha(\alpha-2) \dots (\alpha-2n+2)(\alpha+1)(\alpha+3) \dots (\alpha+2n-1)}{(2n)!} a_0$

odds: $a_{2n+1} = (-1)^n \frac{(\alpha-1)(\alpha-3) \dots (\alpha-2n+1)(\alpha+2)(\alpha+4) \dots (\alpha+2n)}{(2n+1)!} a_1$

$$\text{So } y(x) = a_0 \sum_{n=1}^{\infty} (-1)^n \frac{x(x-2) \dots (x-2n+2)(x+1)(x+3) \dots (x+2n-1)}{(2n)!} x^{2n} \\ + a_1 x + a_1 \sum_{n=1}^{\infty} (-1)^n \frac{(x-1)(x-3) \dots (x-2n+1)(x+2)(x+4) \dots (x+2n)}{(2n+1)!} x^{2n+1}$$

$$= a_0 y_1 + a_1 y_2$$

Under the assumption that α is an integer, we get two different polynomial solutions depending upon if α is even or odd.

α even:

$\alpha = 0$: then all of y_1 except the first term (1) is zero, so $y_1(x) = 1$.

$\alpha = 2$: then the first two nonzero terms of y_1 are

$$1 + (-1 \cdot \frac{2 \cdot 3}{2} x^2) + \sum 0$$

$$\Rightarrow y_1(x) = 1 - 3x^2$$

~~even~~ $\alpha = 2m$ (even) then

$$\alpha(\alpha-2) \dots (\alpha-2n+2) = 2m(2m-2) \dots (2m-2n+2) \\ = 2^n (m(m-1) \dots (m-n+1))$$

$$= \frac{2^n m!}{(m-n)!}$$

$$\text{and } (x+1)(x+3) \dots (x+2n-1) = (2m+1)(2m+3) \dots (2m+2n-1) \\ = \frac{(2m+2n-1)!!}{(2m-1)!!} \quad ((-1)!! = 1)$$

$$\text{or } = \frac{(2m+2n)!}{2^{m+n}(m!)n!} \cdot \frac{2^m m!}{(2m)!}$$

$$= \frac{(2m+2n)! m!}{2^n (m+n)! (2m)!}$$

Recall: $(2n)!! = 2^n n!$

$$(2n-1)!! = \frac{(2n)!}{2^n n!}$$

So if $\alpha = 2m$ then:

$$Y_1 = 1 + \sum_{n=1}^m (-1)^n \frac{2^n m!}{(m-n)!} \cdot \frac{(2m+2n)! m!}{2^n (m+n)! (2m)!} x^{2n}$$

$$= 1 + \frac{[m!]}{(2m)!} \sum_{n=1}^m \frac{(2m+2n)!}{(m-n)! (m+n)! (2m)!} x^{2n} = P_{2m}(x)$$

a polynomial of degree $2m$.

If α is odd, i.e. $\alpha = 2m+1$, then Y_2 becomes a finite polynomial and after a similar computation

$$Y_2 = x + \frac{(m!)^2}{(2m+1)!} \sum_{n=1}^m (-1)^n \frac{(2m+2n+1)!}{(m-n)! (m+n)! (2n+1)!} x^{2n+1}$$

$$= P_{2m+1}(x)$$

e.g. if $\alpha = 1$, $P_1(x) = Y_2 = x$
 $\alpha = 3$, $P_3(x) = Y_2 = x - \frac{5}{3}x^3$

The polynomials $P_n = \begin{cases} P_{2m} & n \text{ even} \\ P_{2m+1} & n \text{ odd} \end{cases}$ are multiples of ~~odd~~

the Legendre Polynomials (scaled so that ~~the~~ $L_n(1) = 1$)

n	$P_n(x)$	$L_n(x)$
0	1	1
1	x	x
2	$1 - 3x^2$	$\frac{1}{2}(3x^2 - 1)$
3	$x - \frac{5}{3}x^3$	$\frac{1}{2}(5x^3 - 3x)$
4	$1 - 10x^2 + \frac{35}{3}x^4$	$\frac{1}{8}(35x^4 - 30x^2 + 3)$

If we consider the function

$$f(x, t) = \frac{1}{\sqrt{1-2xt+t^2}} \quad \text{then}$$

$$f_t = \frac{x-t}{(1-2xt+t^2)^{3/2}}, \quad f_{tt} = \frac{3(x-t)^2}{(1-2xt+t^2)^{5/2}} - \frac{1}{(1-2xt+t^2)^{3/2}}$$

and so if $t=0$:

$$f(x, 0) = \frac{1}{1^{1/2}} = 1 \rightarrow \frac{f}{0!} = 1 = L_0(x)$$

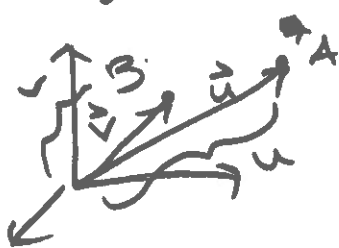
$$f_t(x, 0) = \frac{x}{1^{3/2}} = x \rightarrow \frac{f_t}{1!} = x = L_1(x)$$

$$f_{tt}(x, 0) = \frac{3x^2}{1^{5/2}} - \frac{1}{1^{3/2}} = 3x^2 - 1 \rightarrow \frac{f_{tt}}{2!} = \frac{3x^2 - 1}{2} = L_2(x)$$

and in general $\frac{\frac{\partial^n f}{\partial t^n}(x, 0)}{n!} = L_n(x).$

So $\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} L_n(x) t^n$ (generating function)

We can obtain this generating function when discussing the gravitational potential or Coulomb potential of two point objects. If $\vec{u}, \vec{v}$ are the positions of the two objects and u and v their distances from the origin, then this potential is given by the inverse of their related distance i.e.



$$\frac{1}{|\vec{u} - \vec{v}|}$$

$$\frac{1}{|\vec{u} - \vec{v}|} = \frac{1}{\sqrt{u^2 + v^2 - 2uv \cos \theta}} = \frac{1}{u \sqrt{1 - 2\frac{v}{u} \cos \theta + (\frac{v}{u})^2}}$$

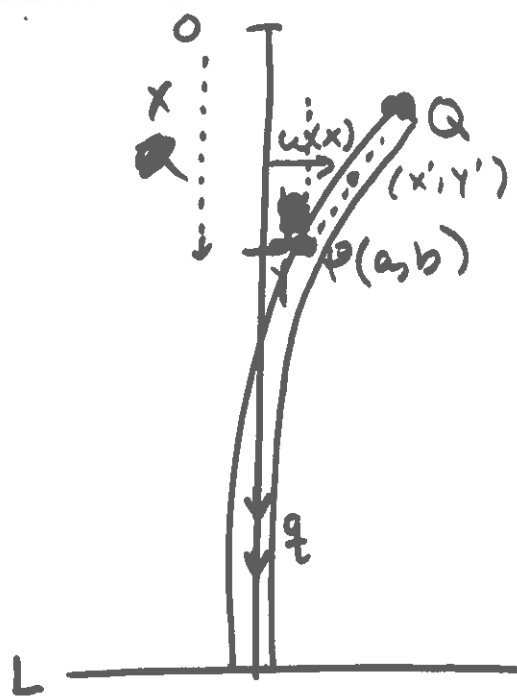
Note:

$$\begin{aligned} |\vec{u} - \vec{v}|^2 &= (\vec{u} - \vec{v}) \cdot (\vec{u} - \vec{v}) = \vec{u} \cdot \vec{u} + \vec{v} \cdot \vec{v} - 2\vec{u} \cdot \vec{v} \\ &= |\vec{u}|^2 + |\vec{v}|^2 - 2\vec{u} \cdot \vec{v} \\ &= u^2 + v^2 - 2uv \cos \theta \end{aligned}$$

$$= \frac{1}{u} \cdot \frac{1}{\sqrt{1 - 2(\frac{v}{u}) \cos \theta + (\frac{v}{u})^2}} \quad \text{with } \begin{aligned} x &= \cos \theta \\ t &= \frac{v}{u} \end{aligned}$$

$$= \frac{1}{u} \sum_{n=0}^{\infty} L_n(\cos \theta) \left(\frac{v}{u}\right)^n = \sum_{n=0}^{\infty} \frac{v^n}{u^{n+1}} L_n(\cos \theta)$$

Self-Buckling Tower



δ - density (volumetric)
 A - cross-sectional area
 g - gravity
 $\rho = \delta A$
 body force $q := \delta g A$
 density

Moment Equilibrium is

$$M = - \int_0^x q (y - w) dx$$

According to elastic ^{beam} theory
 (see derivation of the bending
 equation under
 Euler-Bernoulli beam theory)

$$M = -EI \frac{d^2 w}{dx^2}$$

E - Yang's modulus
 I - second moment of area (A)

$$\Rightarrow EI \frac{d^2 w}{dx^2} = \int_0^x q (y' - w) dx' = q \int_0^x (y' - w) dx'$$

$$\frac{d}{dx} \left[EI \frac{d^2 w}{dx^2} = q \int_0^x (y' - w) dx' \right] \Rightarrow EI \frac{d^3 w}{dx^3} = q \int_0^x 0 - \frac{dw}{dx} dx'$$

$$\Rightarrow EI \frac{d^3 w}{dx^3} = -q x \frac{dw}{dx}$$

$$\text{set } w' = z \Rightarrow EI \frac{d^2 z}{dx^2} + q x z = 0$$

$$\Rightarrow z'' + \frac{q}{EI} x z = 0$$

$$\text{let } k = \sqrt{\frac{q}{EI}} = \sqrt{\frac{\delta g A}{EI}} \Rightarrow z'' + k^2 x z = 0$$

$$\text{or } y'' + k^2 x y = 0 \text{ (Airy equation)}$$

Airy Equation Tower (Self Buckling)

$$y'' + k^2 x y = 0$$

$y'(0) = 0, \quad y(L) = 0$
 $\uparrow$
 $a_1 = 0$

 $k = \sqrt{\frac{9\rho}{EI}}, \quad \rho = \delta A$

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} k^2 a_n x^{n+1} = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=1}^{\infty} k^2 a_{n-1} x^n = 0$$

$$\Rightarrow 2a_2 + \sum_{n=1}^{\infty} [(n+2)(n+1) a_{n+2} + k^2 a_{n-1}] x^n = 0$$

$$\Rightarrow \underline{a_2 = 0} \leftarrow \text{constant solution for third order equation.}$$

$$a_{n+2} = \frac{-k^2 a_{n-1}}{(n+2)(n+1)} \quad n \geq 1$$

$$n=1: a_3 = \frac{-k^2 a_0}{(3)(2)} = -1 \cdot \frac{k^2}{3 \cdot 2} a_0 \quad \begin{matrix} n=2: a_4 = \dots a_1 = 0 \\ n=3: a_5 = \dots a_2 = 0 \end{matrix}$$

$$n=4: a_6 = \frac{-k^2 a_3}{6 \cdot 5} = (-1)^2 \frac{(k^2)^2}{(6 \cdot 5)(3 \cdot 2)} a_0$$

$$n=7: a_9 = \frac{-k^2 a_6}{9 \cdot 8} = (-1)^3 \frac{(k^2)^3}{(9 \cdot 8)(6 \cdot 5)(3 \cdot 2)} a_0$$

$$n=10: a_{12}$$

$$n = 3m-2: a_{3m} = (-1)^m \frac{(k^2)^m}{(3m \cdot (3m-1))(3m-3)(3m-4) \dots (3 \cdot 2)} a_0$$

$$\text{So } y = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_{3n} x^{3n} + \sum_{n=0}^{\infty} a_{3n+1} x^{3n+1} + \sum_{n=0}^{\infty} a_{3n+2} x^{3n+2}$$

$$= a_0 \sum_{n=0}^{\infty} (-1)^n \frac{(k^2)^n}{(3n)(3n-1)(3n-2)\dots(3\cdot 1)} x^{3n}$$

if we use $n!!! = \begin{cases} n(n-3)\dots 3, & n=3k \\ n(n-3)\dots 2, & n=3k+1 \\ n(n-3)\dots 1, & n=3k+2 \end{cases}$

$$= a_0 \sum_{n=0}^{\infty} (-1)^n \frac{k^{2n}}{(3n)!!! (3n-1)!!!} x^{3n}$$

Note: $(3n)!!! = 3n(3n-3)(3n-6)\dots 3 = 3^n (n(n-1)\dots 1) = 3^n n!$

$$(3n-1)!!! = (3n-1)\dots (5)(2)$$

$$= a_0 \sum_{n=0}^{\infty} (-1)^n \frac{k^{2n}}{3^n n! (3n-1)\dots (5)(2)} x^{3n}$$

We have that the column when $y(L) = 0$ is satisfied and

$$\text{so } 0 = a_0 \sum_{n=0}^{\infty} (-1)^n \frac{k^{2n}}{3^n n! (3n-1)\dots (5)(2)} (L)^{3n} \quad \tilde{y}(L)$$

$a_0 \neq 0$ or there is no nontrivial solution. So we must have

$$\sum_{n=0}^{\infty} (-1)^n \frac{k^{2n}}{3^n n! (3n-1)\dots (5)(2)} L^{3n} = 0 \quad \text{or } \tilde{y}(L) = 0$$

where L is the first root of $\tilde{y}$. If we are given a value for k we can find this.

Recall $k = \sqrt{\frac{9P}{EI}}$

What if we don't know k or want to find this for various object at once?

The function

$$J_p(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(p+n+1)} \left(\frac{x}{2}\right)^{2n+p} \quad (1)$$

is called the Bessel Function of order p of the first kind

(it solves $x^2 y'' + xy' + (x^2 - p^2)y = 0$)

It turns out that $y = a_0 \sum_{n=0}^{\infty} (-1)^n \frac{k^{2n}}{3^n n! (3n-1)(3n-4)\dots(5)(2)} x^{3n}$

$$= \tilde{a}_0 x^{1/2} J_{-1/3} \left(\frac{2}{3} k x^{3/2} \right)$$

Proof of (*):

$$\tilde{a}_0 x^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(-\frac{1}{3}+n+1)} \left(\frac{2/3 k x^{3/2}}{2} \right)^{2n-\frac{1}{3}} \quad (1)$$

$$= \tilde{a}_0 x^{1/2} \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{2}{3}\right)^{2n} k^{2n} (x^{3/2})^{2n}}{2^{2n} n! \Gamma(n+\frac{2}{3})} \frac{\left(\frac{2}{3}\right)^{-1/3} k^{-1/3}}{2^{-1/3}} (x^{3/2})^{-1/3}$$

$$= \tilde{a}_0 x^{1/2} \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n} k^{2n} x^{3n}}{2^{2n} n! 3^{2n} (n-\frac{1}{3})(n-\frac{4}{3})\dots\Gamma(\frac{2}{3})} \frac{\left(\frac{2}{3}\right)^{-1/3} k^{-1/3}}{2^{-1/3}} x^{-1/2}$$

↖ constant

$$= \underbrace{(\tilde{a}_0 \cdot \text{constant})}_{a_0} \sum_{n=0}^{\infty} (-1)^n \frac{k^{2n}}{3^n n! (3n-1)(3n-4)\dots(5)(2)} x^{3n} = y$$

so we have $y = \tilde{a}_0 x^{1/2} J_{-1/3} \left(\frac{2}{3} k x^{3/2} \right)$

and so $0 = y(L) = \tilde{a}_0 L^{1/2} J_{-1/3} \left(\frac{2}{3} k L^{3/2} \right)$

So we get buckling when $\frac{2}{3} k L^{3/2}$ is the smallest positive root of $J_{-1/3}(x)$ (i.e. when $J_{-1/3}(\frac{2}{3} k L^{3/2}) = 0$ first)

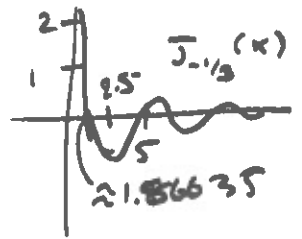
rounded to 5 decimal places this is ≈ 1.86635

thus it will buckle first when $\frac{2}{3} k L^{3/2} = 1.86635$

$$\Rightarrow L^{3/2} = \frac{3Z_1}{2k} \Rightarrow L = \left(\frac{3Z_1}{2k} \right)^{2/3}$$

using $k = \sqrt{\frac{\rho g}{EI}} \Rightarrow L = \left[\frac{3Z_1}{2} \left(\frac{EI}{\rho g} \right)^{1/2} \right]^{2/3}$

$$\approx (1.986) \left(\frac{EI}{\delta A g} \right)^{1/3}$$



where $\rho = \delta A$, δ - volumetric density and A cross-sectional area.

For steel, $E = 2.8 \times 10^7$ psi, $\rho = 0.28$ lb/in³

and so $L \approx (1.986) \left(\frac{2.8 \times 10^7 I}{0.28 \cdot A} \right)^{1/3} \approx 921.8195 \left(\frac{I}{A} \right)^{1/3}$

if the rod is circular $A = \pi r^2$, $I = \frac{1}{4} \pi r^4$

$$\Rightarrow L \approx 921.8195 \left(\frac{\frac{1}{4} \pi r^4}{\pi r^2} \right)^{1/3}$$

$$\approx 580.7099 (r^2)^{1/3}$$

$r = 0.5$ in gives $L \approx 365.8743$ in or 30 ft 6 in

$r = 1.5$ in gives $L \approx 760.9452$ in or 63 ft 5 in

if the rod is hollow (tube)

~~inner~~ $r_{\text{inner}} = 1.25$ in $r_{\text{outer}} = 1.5$ in gives 75 ft 7 in

